Task:

- First built a custom dataset of teacups(25), uno flip cards(26) and hand gestures(26). (the teacups were mostly white with a little pattern)
- Separated the images in class labelled folders, and then uploaded the main folder(in zip format) to google colab (and unzipped the folder)
- Separated the dataset (class wise) into training(62 img) and validation (15 img), rescaled the images pixel from 0-255 to 0-1, and changed it to 1D format.
- We used softmax activation function because we had more than 2 classes
- We compiled the model (I didn't understand this part of the code)
- After the training part was done, i tested on a teacup image from the dataset (i know this part would cause data leakage tho) anyways the result was- [[0.9945446 0.00255767 0.00289773]] , and yeah the predictions was a teacup. [[9.8886222e-01 4.7702648e-04 1.0660758e-02]] when i uploaded a different image with same teacup and background not in dataset, the output was still pretty accurate
- Then i tested an image of a teacup very different from the teacup images i uploaded(bright red) and the answer was `[[0.00372127 0.3888019 0.6074768 ]] Predicted: unoflipcards`

  During this the epochs were = 10, lets try increasing them epochs=20
  `[[0.00423522 0.03800855 0.9577563 ]]`
  `Predicted: unoflipcards`
  Why did increasing epochs made the prediction even worse when the model should be learning better
  Then i decreased epochs to 10 and tested with new images for hand gestures and uno flip cards and it still predicted teacups as the answer
  Then i put an image again from dataset of unoflip cards and it was correct output `[[1.0451565e-02 3.4871304e-05 9.8951352e-01]]`

  For the problem that the model was still predicting teacups for every other class, could be that it never learned other classes that well and it is the first label (0), so then i applied the dropout method, where you basically switch off some neurons and fire again so to make sure that the model is not just memorizing the output.
  After that my new hand gesture image got the output - `[[0.43828782 0.05522745 0.50648475]] unoflipcards`

  `But my new unoflip image got output as [[0.45598295 0.07325361 0.47076347]] as unoflipcards which was the correct output`

Experiment 1: epochs=10, no dropout, overfit, 95% confidently wrong on unseen images
Experiment 2: epochs=20, with dropout, uncertain, basically guessing
Experiment 3: epochs=20, no dropout, confidently wrong on different classes